

Md Raihan Ahmed

SOFTWARE ENGINEER · SECURITY RESEARCHER

Salt Lake City, Utah, USA

✉ u1374605@utah.edu | 🏠 mdrahmed.github.io | 📱 mdrahmed | 🌐 mdrahmed

“Let me help you make the change that you want to see in this world.”

Education

University of Utah

PH.D. IN COMPUTER SCIENCE, GPA: 3.86

Utah, USA

Aug. 2021 - Aug. 2026 (Expected)

University of Utah

MS. IN COMPUTER SCIENCE, GPA: 3.86

Utah, USA

Aug. 2021 - May 2024

Experience

Meta

Menlo Park, California, USA

SOFTWARE ENGINEER INTERN

May 2025 - Aug. 2025

- Identified limitations in manual asset classification processes for securing user data, highlighting significant false positives/negatives due to lack of automation.
- Identify a scalable solution **to improve speed and accuracy.**
- Developed and implemented **machine learning models to automate asset classification.** Then, compared candidate models, selected the best, and deployed it into production.
- Significantly reduced the aggregation process timeline** from months to a fraction of the time.

University of Utah

Utah, USA

RESEARCH ASSISTANT

Aug. 2021 - Current

- Advanced persistent threats (e.g., Stuxnet, Ukrainian power grid attack) exposed vulnerabilities in Industrial Control Systems (ICS).
- Built ICSTracker using C++ & LLVM **to instrument controllers, enabling full activity tracing.**
- Simulated multiple cyberattacks following MITRE ATT&CK framework on testbeds** (Fischertechnik factory, SWaT water system) to validate robustness.
- Traced attack origins in real-time with the tool successfully**, demonstrating ability to enhance ICS security and contributing to peerreviewed publication.

University of Utah

Utah, USA

TEACHING ASSISTANT

- Teaching Assistant for CS 4400-001 - Computer System in Fall-2022 & Fall-2024
- Teaching Assistant for CS 6956-001 - Software and System Security in Spring-2023

Projects

ICSTracker: To Trace back to the source of Intrusion in Industrial IoT Systems.

C++, LLVM, Python

PHD 1ST PROJ

GitHub Aug. 2021 - May 2024

- Created an instrumentation tool with LLVM to trace back to the source of attack.
- Physical process-aware, cross-iteration and cross-domain approach to backtracking intrusions in ICS
- Crafted detailed instrumentation passes to capture critical runtime data from ICS controllers; improved data collection accuracy by 56%, to enable more effective troubleshooting for complex system failures.

Interactive Visualization tool

Javascript

DATA VISUALIZATION COURSE

GitHub Aug. 2022 - Dec. 2022

- Formulated project is to build an interactive tool for video games.
- Built website with javascript to explore video game sales by year and platform.

Fuzzing XPDF-4.05

LLVM, C++

APPLIED S/W SECURITY TEST. COURSE

Git1 Git2 Aug. 2022 - Dec. 2022

- Extended fuzzing campaign with static analysis (control flow + IR inspection) to guide fuzzing input generation.
- Applied program verification tools (Z3, CBMC) to validate discovered crash paths.
- Improved open-source robustness by contributing bug reports and patches.

Rollback attack in Cars

CYBER-PHYS SYS & IOT SECURITY COURSE

Arduino, C++
Github Aug. 2022 - Dec. 2022

- Formulated project was to build a device to perform cyberattack on modern cars.
- Built a rollback device with Arduino to access cars without using keys.

Remote control robot built with Arduino

A PERSONAL COOL PROJECT

Arduino, C++
Demo May 2021

- Used arduino uno to build this robot and controlling it with Bluetooth

Publications

IEEE/IFIP International Conference on Dependable Systems and Networks

Naples, Italy

ICSTRACKER: BACKTRACKING INTRUSIONS IN MODERN INDUSTRIAL CONTROL SYSTEMS (1ST AUTHOR)

paper Apr. 2025

- Recovered program semantics, reconstructed data dependencies, and linked controller operations to OS- level events to show the cyber-physical attacks in ICS/CPS setting.

Re-design Industrial Control Systems with Security (RICSS)

Utah, USA

CONTEXT-AWARE INTRUSION DETECTION IN INDUSTRIAL CONTROL SYSTEMS (1ST AUTHOR)

paper Oct. 2024

- Developed a provenance-based method with Linear Temporal Logic (LTL) to enhance ICS attack detection in information security.

Biointerface Research in Applied Chemistry

Bangladesh

OVARIAN CANCER SUBSTANTIAL RISK FACTOR ANALYSIS BY MACHINE LEARNING: A LOW INCOMING COUNTRY (1ST)

paper Jul. 2020

- Analyzed ovarian cancer data using various machine learning algorithms to identify significant risk factors

BMC bioinformatics

Bangladesh

MACHINE LEARNING TO REVEAL AN ASTUTE RISK PREDICTIVE FRAMEWORK FOR GYNECOLOGIC CANCER AND ITS

GitHub paper Apr. 2021

IMPACT ON WOMEN PSYCHOLOGY: BANGLADESHI PERSPECTIVE (B.Sc PROJ, 2ND AUTHOR)

- Developed a predictive algorithm using machine learning to assess the risk of cervical and ovarian cancer associated with stress

Electrical, Computer and Communication Engineering (ECCE)

Bangladesh

A COMPARATIVE ANALYSIS OF TRADITIONAL AND MODERN DATA COMPRESSION SCHEMES FOR LARGE

paper Feb. 2019

MULTIDIMENSIONAL EXTENDIBLE ARRAY (5TH AUTHOR)

- Compared traditional and modern data compression schemes for multi-dimensional data and concluding that the Extendible Array Based Compression Scheme (EaCRS) is most efficient

Elsevier

Bangladesh

A BIOINFORMATICS APPROACH FOR IDENTIFICATION OF THE CORE ONTOLOGIES AND SIGNATURE GENES OF

paper Jul. 2020

PULMONARY DISEASE AND ASSOCIATED DISEASES (2ND AUTHOR)

- Spotted out significant common genes and proteins among COPD, DM, CR, IHD, IS, TB, and OB diseases, to suggest drug signatures.

Journal of Proteins and Proteomics

Bangladesh

PROTEIN INTERACTION NETWORK AND DRUG DESIGN OF STOMACH CANCER AND ASSOCIATED DISEASE: A

paper Dec. 2020

BIOINFORMATICS APPROACH (1ST AUTHOR)

- Identified common genes among seven diseases (OB, SC, CC, PC, PRC, LK, MD) to analyze their biological and genetic network.

Skills

Dev. & Deployment Tools

LLVM Compiler Tools, GitHub, Docker

Programming Lang.

Python, C/C++, Javascript, C#, JAVA, PHP

Frameworks

MITRE ATT&CK, Cyber Kill Chain, React.js, Node.js, Express.js, Android Studio, Dot(.) Net

Data Science & ML

Tensorflow, NumPy, scikit-learn, Pandas, Matplotlib

Database

Oracle SQL, MySQL

Security Tools

AFL++, Z3, CBMC, Snort, Suricata, OSS-Fuzz, Provenance Tracking

Data Analysis Software

Matlab, Google Colab, Jupyter, Anaconda

Cloud & Container Security

Docker, Kubernetes

Cyber Defense & Security Engineering

Threat Hunting, Threat Intelligence, Incident Response, Vulnerability Management, Threat Modeling

Interests

Security, Privacy, Machine learning, Artificial Intelligence